

FPGA Based Real Time Object Detection on PYNQ-Z2

PROJECT REPORT

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**MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY
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BONAFIDE CERTIFICATE

This is to certify that the project work entitled “**FPGA BASED REAL TIME OBJECT DETECTION ON PYNQ-Z2**” is a bonafide work done by **BATHREESH.M [REGISTER NO: 23TC0216], GOKULAKANNAN.S [REGISTER NO: 23TC0236], JAGADEESH.G [REGISTER NO: 23TC0246]** in partial fulfillment of the requirement for the award of B. Tech degree in **ELECTRONICS AND COMMUNICATION ENGINEERING** by Pondicherry University during the academic year 2023 – 2027.

HEAD OF THE DEPARTMENT

PROJECT GUIDE

Viva-Voce Examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

Certified that the Report entitled “**FPGA BASED REAL TIME OBJECT DETECTION ON PYNQ-Z2**” is the bonafide record of independent work done by **BATHREESH.M [REGISTER NO: 23TC0216], GOKULAKANNAN.S [REGISTER NO: 23TC0236], JAGADEESH.G [REGISTER NO: 23TC0246]**, for the award of B. Tech degree in **ELECTRONICS AND COMMUNICATION ENGINEERING** under the supervision of **Dr. S. ARUNMOZHI**. Certified further that the work reported herein does not from part of any other thesis or dissertation on the basis of which a degree or award was conferred earlier.

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PROJECT GUIDE

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We thank the Almighty for blessing us with such wonderful people and for being with us always



SUSTAINABLE DEVELOPMENT GOALS (SDGs) MAPPING

Title : **FPGA-Based Real-Time Object Detection on PYNQ-Z2**
SDG Goal -9 : **Industry, Innovation And Infrastructure**



This project supports SDG Goal 9 because it combines embedded hardware, intelligent software and real-time system design in a way that encourages innovation and practical infrastructure development. The PYNQ-Z2 platform represents an accessible edge-computing environment where hardware acceleration, image processing and machine learning can work together. The project demonstrates how modern embedded platforms can be used to build efficient and low-power intelligent systems that are suitable for academic research, industrial prototypes and future smart applications. In a broader sense, the work contributes to the idea of innovation-driven infrastructure because it shows how a compact FPGA-based board can support a real-time vision pipeline that is normally expected from larger and more expensive systems.

ABSTRACT

This project presents an FPGA-based real-time object detection system implemented on the PYNQ-Z2 platform using a hardware-software co-design approach. The system combines the Processing System (PS), which runs on the ARM Cortex-A9 processor, with the Programmable Logic (PL), which is used for high-speed data movement through AXI DMA. Live video is captured from a USB camera, enhanced using CLAHE to improve visibility in dark or backlit scenes, and then processed using YOLOv3-tiny for object detection. The model identifies common indoor objects such as laptops, persons, chairs, bottles, cups, keyboards and mice with bounding boxes and confidence scores.

A special feature of the system is that the full real-time detection pipeline is executed in a compact embedded environment. The FPGA block design handles communication between the processor and the programmable logic, while the Python application running inside Jupyter Notebook manages camera capture, preprocessing, inference and result visualization. Because the available DMA buffer is limited, the design uses a smaller grayscale thumbnail for DMA transfer while retaining the full enhanced frame for object detection. This allows the project to demonstrate efficient PS-PL interaction without sacrificing the real-time detection workflow.

The proposed work highlights how embedded AI can be built using affordable development hardware when careful attention is given to data movement, model selection and image preprocessing. The results show that the system can successfully detect objects in live frames and display annotated outputs in near real time. Therefore, the project serves as a useful example of practical FPGA-based intelligent vision design and can be extended further for surveillance, monitoring and edge-computing applications.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	BONAFIED CERTIFICATE	ii
	DECLARATION	iii
	ACKNOWLEDGEMENT	iv
	SDG GOALS MAPPING	v
	ABSTRACT	vi
	LIST OF FIGURES	ix
	LIST OF TABLES	1
1	INTRODUCTION	2
1.1	PRELUDE	1
1.2	LITERATURE SURVEY	2
1.3	PROBLEM STATEMENT	3
1.4	OUTLINE	3
1.5	MOTIVATION	4
1.6	ORGANIZATION OF THE REPORT	4
2	FPGA-BASED REAL-TIME OBJECT DETECTION METHODOLOGY	5
2.1	AIM	5
2.1.1	Objectives	6
2.2	SCOPE	7
2.3	SYSTEM ARCHITECTURE	8
2.4	HETEROGENEOUS PS-PL ARCHITECTURE	10
2.5	PYNQ-Z2 HARDWARE OVERVIEW	11
2.6	YOLOv3-TINY OVERVIEW	13
2.7	AXI DMA COMMUNICATION	15
2.8	PROPOSED DATA FLOW	16

3	SYSTEM IMPLEMENTATION	19
3.1	HARDWARE SETUP	20
3.2	SOFTWARE ENVIRONMENT	22
3.3	VIVADO IP BLOCK DESIGN	24
3.4	OVERLAY LOADING PROCESS	28
3.5	DMA TRANSFER AND OPTIMIZATION	30
3.6	IMAGE PREPROCESSING USING CLAHE	32
3.7	YOLOV3-TINY DETECTION PIPELINE	34
3.8	LABEL REMAPPING TECHNIQUE	36
4	RESULTS AND PERFORMANCE ANALYSIS	38
4.1	OVERVIEW	38
4.2	DETECTION RESULTS	39
4.2.1	object detection frame analysis	40
4.2.2	detection accuracy analysis	42
4.2.3	fps performance analysis	43
4.2.4	dma performance analysis	44
4.3	CHALLENGES AND SOLUTIONS	45
4.4	DISCUSSION	48
5	CONCLUSION	49
5.1	CONCLUSION	49
5.2	FUTURE SCOPE	50
	REFERENCES	52

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
2.1	System Data Flow	9
2.2	Proposed System Architecture	9
2.3	PS–PL Cooperation Concept	10
2.4	Heterogeneous PS–PL Co-Design Architecture	11
2.5	PYNQ-Z2 Development Board	12
2.6	AXI DMA Communication Flow	16
2.7	Proposed System Flow	17
2.8	Proposed Data Flow Diagram	18
3.1	Hardware Setup	21
3.2	Hardware Connection Flow	22
3.3	Software Environment Workflow	23
3.4	Vivado Block Design	24
3.5	ZYNQ Processing System	25
3.6	AXI DMA IP	25
3.7	AXI Interconnect	26
3.8	AXI4-Stream FIFO	27
3.9	Processor System Reset	27
3.10	XLConcat Block	28
3.11	Overlay Loading Process	29
3.12	DMA Transfer Flow	31
3.13	CLAHE Processing Flow	33
3.14	YOLOv3-tiny Detection Pipeline	35
3.15	Label Remapping Process	36

4.1	Detection Output Frame-1	40
4.2	Detection Output Frame-2	41
4.3	Detection Output Frame-3	41
4.4	FPS Performance	43

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
2.1	PYNQ-Z2 Hardware Specifications	14
2.2	YOLOv3-tiny Configuration Parameters	15
2.3	DMA Data Transfer Comparison	17
2.4	Functional Stages of Proposed Data Flow	19
3.1	Hardware Components Used	22
3.2	Software Tools and Frameworks	25
3.3	Overlay Loading Components	31
3.4	DMA Transfer Comparison	33
3.5	CLAHE Parameters Used	35
3.6	Detection Parameters	36
3.7	Example Label Remapping Rules	38
4.1	Frame Detection Summary	43
4.2	Detection Parameter Analysis	44
4.3	FPS Analysis	45
4.4	DMA Transfer Analysis	46
4.5	Challenges and Solutions Summary	49

CHAPTER 1

INTRODUCTION

1.1 PRELUDE

Recent advancements in Artificial Intelligence (AI) and embedded systems have significantly improved computer vision applications. Among them, object detection plays an important role in identifying and locating objects in images and video streams. Real-time object detection is widely used in surveillance, autonomous vehicles, robotics, healthcare, and industrial automation where fast processing and accuracy are essential.

Traditional object detection methods generally depend on CPUs and GPUs, which provide high performance but often consume more power and introduce latency in embedded systems. These limitations create the need for efficient hardware platforms capable of supporting AI applications with lower power consumption and faster processing.

Field Programmable Gate Arrays (FPGAs) provide parallel processing capability and hardware flexibility, making them suitable for real-time embedded applications. The PYNQ-Z2 platform integrates an ARM Processing System (PS) and FPGA Programmable Logic (PL), enabling hardware–software co-design.

In this project, a real-time object detection system is developed using PYNQ-Z2, CLAHE image enhancement, AXI DMA communication, and YOLOv3-tiny. Experimental results demonstrate real-time detection of objects such as laptops, bottles, chairs, and persons.

1.2 LITERATURE SURVEY

Real-time object detection has gained significant attention in embedded applications including surveillance, robotics, healthcare, and intelligent transportation systems.

In [1], YOLO introduced a single-stage object detection framework that performs localization and classification simultaneously, achieving faster detection speed.

In [2], YOLOv3 improved detection performance using the Darknet-53 backbone and multi-scale prediction. However, its complexity limited deployment on embedded platforms.

In [3], YOLOv3-tiny was proposed as a lightweight version suitable for resource-constrained devices while maintaining acceptable detection accuracy.

In [4], FPGA-based acceleration techniques demonstrated reduced latency and lower power consumption compared with CPU-based systems.

In [5], hardware–software co-design using Zynq architecture showed improved throughput by distributing tasks between Processing System (PS) and Programmable Logic (PL).

In [6], AXI DMA communication enabled high-speed data transfer between PS and PL with lower processor overhead.

In [7], CLAHE-based image enhancement improved object visibility under poor lighting conditions.

The literature indicates that lightweight neural networks, FPGA acceleration, and optimized communication techniques contribute toward efficient embedded AI implementation.

1.3 PROBLEM STATEMENT

Real-time object detection systems require continuous image processing with high accuracy and low latency. Traditional CPU and GPU implementations provide good performance but consume higher power and may not be suitable for embedded environments.

Embedded platforms face challenges such as limited memory, computational constraints, and communication overhead. In FPGA systems, transferring large image frames through DMA can create memory limitations and reduce processing efficiency. Environmental conditions such as poor lighting and shadows may also affect detection accuracy.

Therefore, an optimized FPGA-based object detection system is required to provide efficient communication, lower latency, and improved performance. This project addresses these challenges using PYNQ-Z2, AXI DMA, CLAHE, and YOLOv3-tiny.

1.4 OUTLINE

This project presents an FPGA-based real-time object detection system implemented on the PYNQ-Z2 platform using hardware–software co-design. Video frames captured through a USB camera are processed using image enhancement, DMA communication, and object detection stages.

AXI DMA provides communication between PS and PL, while CLAHE improves image quality under poor lighting conditions. YOLOv3-tiny performs object detection because of its lightweight structure and suitability for embedded systems.

The proposed approach integrates hardware acceleration, communication optimization, and deep learning techniques for efficient real-time embedded AI applications.

1.5 MOTIVATION

The motivation for this project comes from the growing demand for intelligent systems capable of performing real-time object detection in embedded environments such as surveillance, robotics, and industrial automation.

Traditional systems require powerful CPUs and GPUs, which consume more power and increase hardware cost. FPGA platforms provide an alternative through parallel processing and hardware acceleration.

PYNQ-Z2 combines ARM Processing System and FPGA Programmable Logic on a single platform, enabling hardware–software co-design. This project also provides practical exposure to FPGA design, computer vision, AXI DMA communication, and embedded AI implementation.

1.6 ORGANIZATION OF THE REPORT

- | | |
|-------------------|---|
| Chapter 1: | Introduces the project, literature survey, problem statement, motivation, and report organization. |
| Chapter 2: | Describes methodology including system architecture, PS–PL design, PYNQ-Z2, YOLOv3-tiny, DMA communication, and data flow. |
| Chapter3: | Explains implementation details including hardware setup, software tools, Vivado block design, CLAHE, DMA transfer, and detection pipeline. |
| Chapter 4: | Presents detection results, FPS analysis, DMA performance, and implementation observations. |
| Chapter 5: | Summarizes conclusions and future enhancements. |

CHAPTER 2

FPGA-BASED REAL-TIME OBJECT DETECTION METHODOLOGY

2.1 AIM

The primary aim of this project is to design and implement an FPGA-Based Real-Time Object Detection System on the PYNQ-Z2 platform using a hardware–software co-design approach. The proposed system aims to utilize both the Processing System (PS) and Programmable Logic (PL) available in the Xilinx Zynq architecture to achieve efficient real-time object detection with reduced processing latency and improved resource utilization.

The project focuses on developing an embedded vision system capable of capturing live video frames through a USB camera and processing them for object identification in real time. The system integrates image preprocessing, high-speed communication mechanisms, and deep learning-based object detection techniques to provide efficient performance on a resource-constrained FPGA platform.

To establish communication between the ARM processor and FPGA fabric, AXI Direct Memory Access (DMA) is used for transferring image data between the Processing System and Programmable Logic. The project also aims to demonstrate effective hardware–software interaction through PS–PL communication while maintaining efficient data movement.

The object detection process utilizes the YOLOv3-tiny model because of its lightweight structure and suitability for embedded applications. The system aims to detect multiple objects such as laptops, bottles, keyboards, chairs, cups, and persons while displaying bounding boxes and confidence values on the output image.

In addition, the project incorporates image enhancement techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve object visibility in dark or backlit environments. By integrating hardware acceleration, image preprocessing, and lightweight deep learning techniques, the proposed work aims to provide a low-power and efficient real-time object detection framework suitable for future intelligent embedded applications.

2.1.1 Objectives

The primary objectives of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 project are to develop an efficient embedded object detection system using hardware–software co-design and to demonstrate real-time communication between the Processing System (PS) and Programmable Logic (PL). The project focuses on achieving faster processing performance, reduced latency, and improved resource utilization using FPGA technology.

The specific objectives of this project are as follows:

1. To implement PS–PL communication using AXI DMA

Develop a communication framework between the ARM Processing System and FPGA Programmable Logic using AXI Direct Memory Access (DMA) to enable efficient high-speed data transfer.

2. To design a hardware–software co-design architecture

Utilize both the Processing System (PS) and Programmable Logic (PL) available in the PYNQ-Z2 platform to distribute software and hardware tasks effectively.

3. To capture real-time video input from a USB camera

Acquire continuous image frames through a USB camera and process them for object detection applications.

4. To improve image quality using preprocessing techniques

Apply Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance image contrast and improve visibility under poor lighting and backlit conditions.

5. To implement lightweight object detection using YOLOv3-tiny

Deploy the YOLOv3-tiny model for real-time object detection because of its lower computational complexity and suitability for embedded systems.

6. To optimize data transfer through DMA communication

Reduce memory overhead by transmitting optimized grayscale thumbnail data through DMA while maintaining full-frame processing for object detection.

7. To perform object detection and classification in real time

Detect and classify multiple objects such as persons, laptops, bottles, keyboards, chairs, and cups by generating bounding boxes and confidence scores.

8. To evaluate system performance under real-time conditions

Analyze parameters such as detection performance, communication efficiency, FPS, and system behavior under different environmental conditions.

9. To demonstrate FPGA-based acceleration for embedded AI applications

Show the advantages of FPGA platforms for implementing intelligent vision systems with lower latency and efficient resource utilization.

These objectives collectively aim to develop a practical and efficient FPGA-based object detection system suitable for future real-time embedded AI applications.

2.2 SCOPE

The scope of this project extends into the fields of embedded Artificial Intelligence, FPGA computing, and real-time computer vision systems, where efficient object detection with low latency and reduced power consumption is increasingly important. The project focuses on implementing a real-time object detection system using the PYNQ-Z2 FPGA platform through a hardware–software co-design approach.

The proposed system utilizes the Processing System (PS) and Programmable Logic (PL) available in the Xilinx Zynq architecture. The ARM processor performs software-related operations such as image acquisition, preprocessing, and object detection, while FPGA logic supports communication and high-speed data transfer. This heterogeneous architecture improves processing efficiency compared with software-only systems.

The scope of the project includes implementing AXI DMA communication for efficient data transfer between PS and PL. The work also investigates communication optimization and memory-efficient techniques for embedded environments.

To improve object detection under challenging conditions, CLAHE preprocessing is applied to enhance image contrast and visibility in low-light and backlit environments. The system utilizes YOLOv3-tiny, a lightweight deep learning model suitable for embedded applications, for detecting multiple object categories with lower computational complexity.

The proposed system can be extended to various real-world applications such as:

- Smart surveillance systems
- Intelligent traffic monitoring
- Autonomous and assisted vehicles
- Industrial automation
- Robotics and machine vision
- Smart healthcare systems
- Edge AI and IoT applications

Furthermore, this work provides a foundation for future research in FPGA-based deep learning acceleration, embedded AI optimization, and hardware-assisted computer vision systems. The project opens opportunities for integrating advanced object detection models, dedicated FPGA accelerators, and real-time intelligent decision-making systems in future implementations.

2.3 SYSTEM ARCHITECTURE

The system architecture of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 is based on a hardware–software co-design approach, where the Processing System (PS) and Programmable Logic (PL) cooperate to perform real-time object detection efficiently. The architecture integrates image acquisition, preprocessing, DMA communication, FPGA interaction, object detection, and output visualization.

The process begins with a USB camera, which continuously captures live video frames. These frames are received by the ARM-based Processing System on the PYNQ-Z2 board and forwarded for preprocessing.

To improve image quality, Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied. This enhancement technique improves local contrast and visibility under low-light and uneven illumination conditions, helping improve object detection performance.

Communication between the Processing System and Programmable Logic is established using AXI Direct Memory Access (DMA). Since transferring complete image frames creates memory limitations, an optimized grayscale thumbnail image is transferred through DMA while preserving the original frame for object detection. This optimization reduces memory usage and improves communication efficiency.

Within the Programmable Logic section, image data passes through AXI4-Stream FIFO and communication modules, enabling successful PS–PL interaction and validating hardware communication.

The enhanced image is then processed using the YOLOv3-tiny model implemented through OpenCV DNN. The model predicts object locations and generates bounding boxes, confidence scores, and labels. To improve output quality, Non-Maximum Suppression (NMS) removes duplicate detections, while label remapping improves prediction consistency.

Finally, the processed output image with detected objects and labels is displayed. The

proposed architecture demonstrates the integration of image enhancement, FPGA communication, and lightweight AI techniques for real-time embedded object detection.

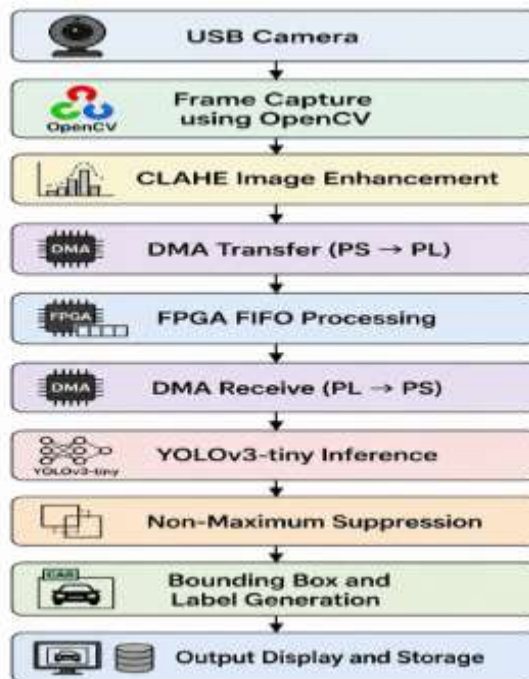


Figure 2.1 System Data Flow

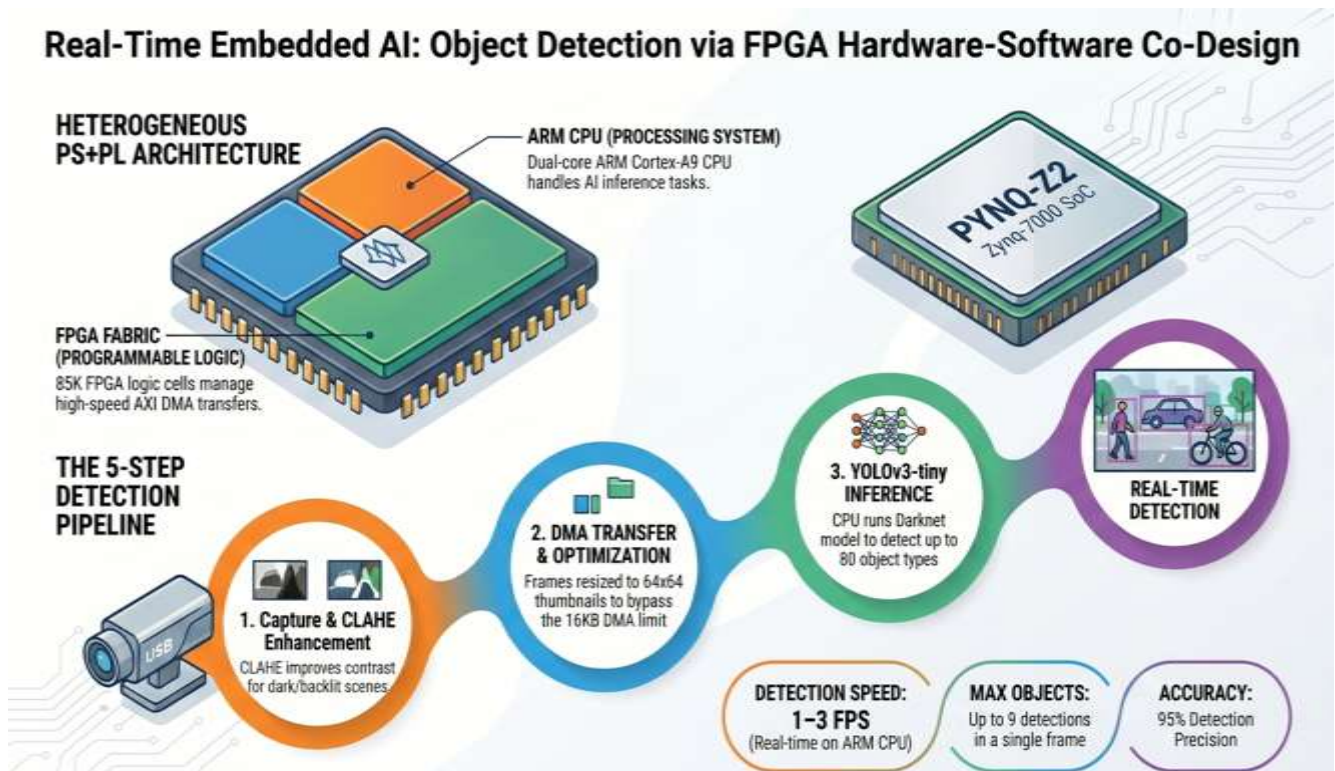


Figure 2.2 Proposed System Architecture

2.4 HETEROGENEOUS PS+PL ARCHITECTURE

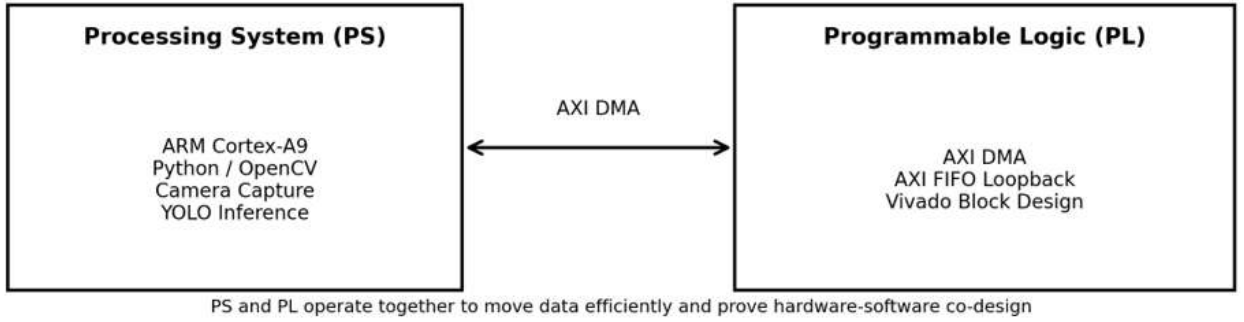


Figure 2.3 Conceptual view of PS and PL cooperation in the proposed system.

The proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system uses a heterogeneous Processing System (PS) and Programmable Logic (PL) co-design architecture to perform real-time object detection efficiently. The PYNQ-Z2 board, based on the Xilinx Zynq-7000 SoC, integrates a dual-core ARM processor and FPGA logic on a single platform. This architecture enables hardware and software components to work together, providing improved performance and lower latency.

In a heterogeneous system, tasks are distributed according to their functionality. Software-related operations are executed in the Processing System, while communication and hardware operations are managed in the Programmable Logic section. This approach combines software flexibility with FPGA parallel processing capability.

In the proposed implementation, the Processing System (PS) performs image acquisition, preprocessing, OpenCV execution, and YOLOv3-tiny inference. The ARM processor captures video frames from the USB camera and applies CLAHE enhancement to improve image visibility before object detection.

The Programmable Logic (PL) handles communication-related operations using AXI DMA, AXI4-Stream FIFO, and supporting hardware modules. These components enable high-speed communication between PS and PL while reducing processor overhead.

Communication between PS and PL is established through the AXI protocol, where optimized image data is transferred through DMA channels. To reduce memory limitations, a grayscale thumbnail image is transferred instead of the complete frame.

One major advantage of the PS–PL architecture is parallel execution. While the Processing System performs AI inference, the Programmable Logic handles communication tasks simultaneously, improving throughput and reducing latency.

The PS–PL co-design approach also supports future expansion by allowing additional FPGA accelerators and custom hardware modules to be integrated into the existing architecture.

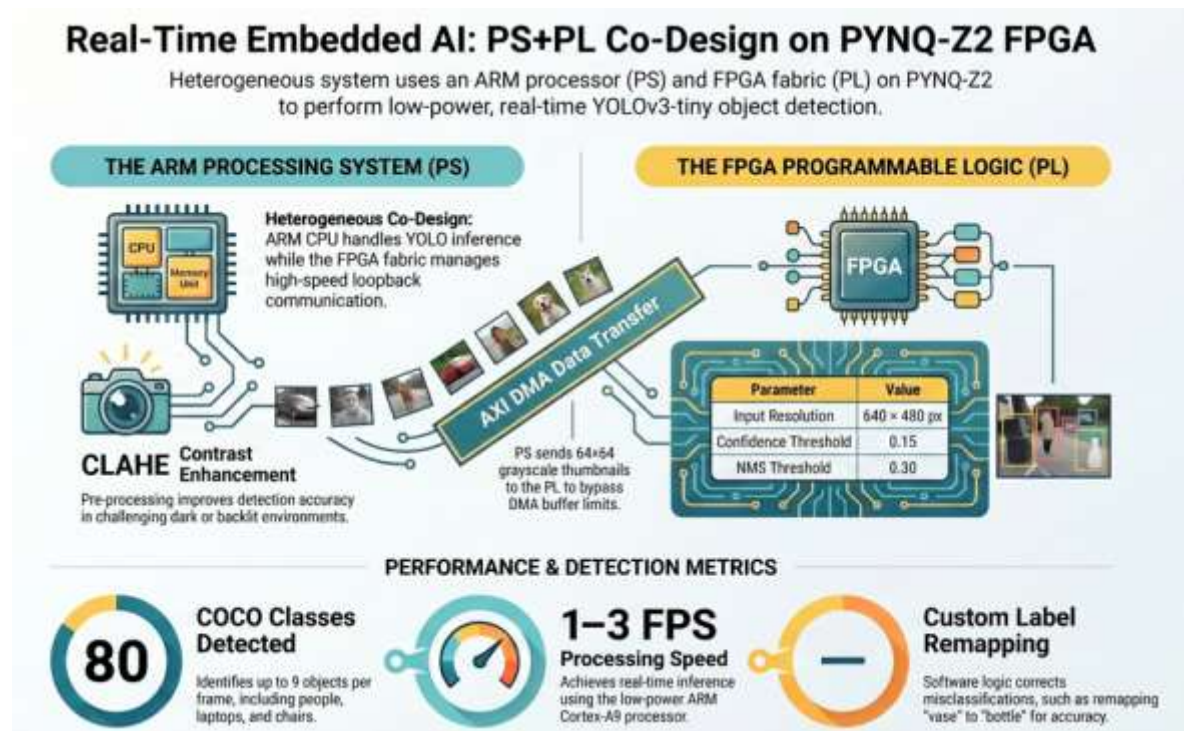


Figure 2.4 Heterogeneous PS–PL Co-Design Architectur

2.5 PYNQ-Z2 HARDWARE OVERVIEW

The PYNQ-Z2 is an FPGA-based development board designed for embedded system applications, hardware acceleration, and machine learning implementation. It is built on the Xilinx Zynq-7000 System-on-Chip (SoC) architecture, which integrates an ARM-based Processing System (PS) and FPGA-based Programmable Logic (PL) within a single device. This integrated architecture creates a heterogeneous platform capable of performing software and hardware operations simultaneously.

A major advantage of the PYNQ-Z2 platform is its support for hardware–software co-design, where computational tasks are distributed between software execution and FPGA hardware resources. This approach improves processing efficiency, reduces latency, and enables parallel operation.

The board used in this project contains the Xilinx Zynq-7020 device, which includes a dual-core ARM Cortex-A9 processor and programmable FPGA logic. The ARM processor executes software operations such as Python applications, image processing algorithms, and object detection functions, while FPGA resources handle communication and hardware-related tasks.

The PYNQ-Z2 board provides several interfaces and peripheral resources including USB ports, HDMI connectivity, Ethernet support, GPIO interfaces, onboard memory resources, LEDs, and push buttons. These features make the platform suitable for rapid prototyping and real-time embedded application development.

The board also supports the PYNQ framework, which allows FPGA hardware to be controlled using Python programming and Jupyter Notebook environments. This simplifies development by enabling interaction with hardware resources through high-level programming methods.

In the proposed system, the PYNQ-Z2 serves as the primary platform for image acquisition, DMA communication, FPGA interaction, and real-time object detection implementation. The heterogeneous architecture enables efficient cooperation between software and hardware modules for embedded AI applications.

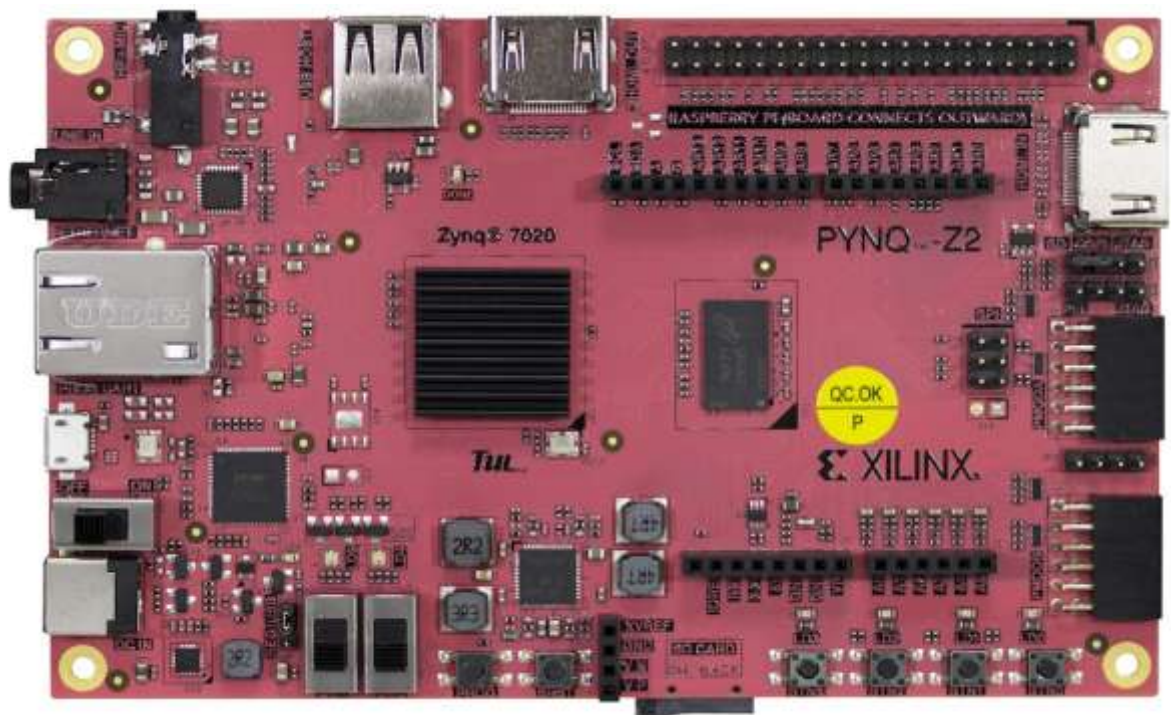


Figure 2.5 PYNQ-Z2 Development Board

Parameter	Specification
FPGA Device	Xilinx Zynq-7020
Processing System	Dual-core ARM Cortex-A9
Programmable Logic	FPGA Fabric
Logic Cells	Approximately 85K
Block RAM	560 KB
DSP Slices	220
Memory	512 MB DDR3
Storage	Micro SD Card
USB Ports	USB Host and USB OTG
Video Interface	HDMI Input / Output
Connectivity	Ethernet
Programming Environment	PYNQ, Vivado, Python

Table 2.1 PYNQ-Z2 Hardware Specifications

2.6 YOLOv3-tiny OVERVIEW

YOLO (You Only Look Once) is a deep learning-based object detection algorithm used to identify and locate objects in images and video frames. Unlike traditional methods, YOLO performs localization and classification simultaneously using a single neural network, enabling faster detection suitable for real-time applications.

YOLOv3-tiny is a lightweight version of YOLOv3 designed for embedded and resource-constrained systems. It reduces network complexity while maintaining acceptable detection accuracy and speed.

The model uses the Darknet framework and predicts object labels, confidence scores, and bounding box coordinates in a single forward pass.

In this project, YOLOv3-tiny is selected because it offers a balance between speed, memory usage, and computational efficiency, making it suitable for the PYNQ-Z2 platform. The model uses pre-trained weights from the COCO dataset, which contains 80 object classes.

Experimental testing successfully detected objects such as laptops, bottles, chairs,

keyboards, cups, and persons, demonstrating the suitability of YOLOv3-tiny for embedded real-time object detection.

During experimental testing, the proposed system successfully detected objects such as:

- Person
- Laptop
- Bottle
- Chair
- Cup
- Keyboard
- Mouse

The object detection process begins by resizing image frames and converting them into a blob representation suitable for neural network input. The YOLOv3-tiny model then processes the image and predicts multiple candidate objects. Each prediction includes:

- Bounding box coordinates
- Object confidence score
- Class probability values

After detection, Non-Maximum Suppression (NMS) is applied to eliminate duplicate overlapping predictions and improve output quality. Additional label remapping logic is also used in the implementation to reduce common misclassification issues.

The use of YOLOv3-tiny enables the system to achieve real-time object detection performance while maintaining computational efficiency. Due to its lightweight structure and faster inference speed, the model is highly suitable for embedded AI applications involving FPGA platforms.

Parameter	Value
Model	YOLOv3-tiny
Framework	Darknet / OpenCV DNN
Input Resolution	416×416
Dataset	COCO
Number of Classes	80
Confidence Threshold	0.15
NMS Threshold	0.30

Table 2.2 YOLOv3-tiny Configuration Parameters.

2.7 AXI DMA COMMUNICATION

Efficient communication between hardware and software modules is essential in FPGA-based embedded systems. In the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system, communication between the Processing System (PS) and Programmable Logic (PL) is established using AXI Direct Memory Access (AXI DMA). AXI DMA enables high-speed data transfer between software and FPGA hardware with minimal processor involvement.

The AXI (Advanced eXtensible Interface) protocol supports communication between the ARM processor and FPGA logic within the Xilinx Zynq architecture. Unlike conventional processor-controlled transfer methods, DMA moves data directly between memory and hardware modules, reducing processor overhead and improving system performance.

In the proposed implementation, image frames captured from the USB camera are initially processed in the Processing System and then transferred through DMA channels for PS–PL communication.

During implementation, complete image frames with a resolution of $640 \times 480 \times 3$ created memory limitations and communication overhead. To overcome this issue, an optimized 64×64 grayscale thumbnail image was transferred through AXI DMA instead of the complete frame. This reduced the transfer size from approximately 921,600 bytes to 4096 bytes, improving communication efficiency.

The transferred image data passes through AXI4-Stream FIFO modules within the Programmable Logic section and performs loopback communication to verify successful PS–PL interaction.

The implementation demonstrates that AXI DMA communication enables efficient data transfer and supports real-time operation in FPGA-based embedded systems.

The use of AXI DMA offers several advantages:

- Reduces processor overhead
- Enables high-speed communication
- Supports efficient PS–PL interaction
- Improves system throughput
- Reduces communication latency
- Allows parallel execution of operations

The successful implementation of AXI DMA communication demonstrates the capability of hardware–software co-design for embedded AI systems and forms an important component of the proposed object detection architecture.

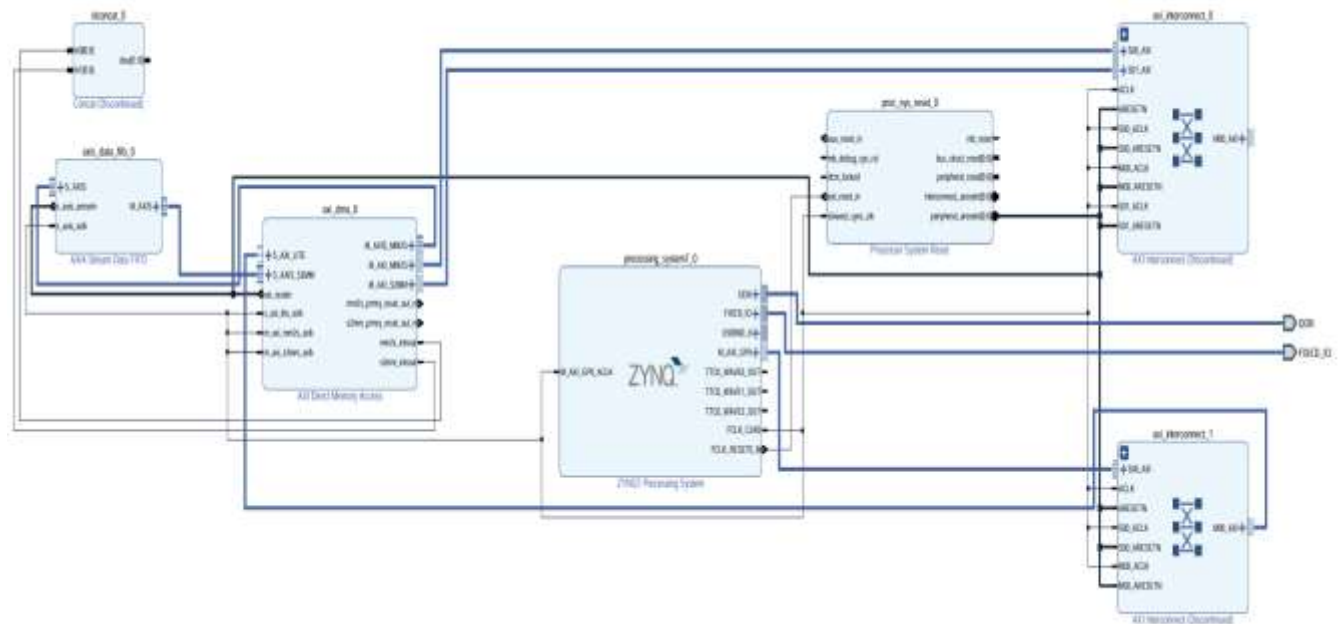


Figure 2.6 AXI DMA Communication Flow

Parameter	Original Image	Optimized Image
Resolution	640 × 480 × 3	64 × 64 × 1
Data Size	921,600 bytes	4,096 bytes
Transfer Type	Full Frame	Thumbnail
DMA Efficiency	Lower	Higher

Table 2.3 DMA Data Transfer Comparison

2.8 PROPOSED DATA FLOW

The proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system follows a structured data flow architecture to achieve efficient communication and real-time object detection. The workflow is designed using a hardware–software co-design approach, where image data passes through multiple stages before generating the final output.

The process begins with a USB camera, which continuously captures live image frames and sends them to the Processing System (PS) of the PYNQ-Z2 platform. The captured frames are initially processed using OpenCV libraries.

Before object detection, CLAHE preprocessing is applied to improve image contrast and visibility under low-light conditions. This enhancement improves image quality and supports better object detection performance.

After preprocessing, communication between the Processing System and Programmable Logic is established using AXI DMA. Since complete image frames create memory limitations, a reduced grayscale thumbnail image is transferred through DMA channels to improve communication efficiency.

Within the Programmable Logic (PL) section, image data passes through AXI4-Stream FIFO modules that provide buffering and loopback functionality, verifying successful communication between hardware and software modules.

After communication, the original enhanced image frame is processed using the YOLOv3-tiny model implemented through OpenCV DNN. The model generates object predictions with labels, confidence values, and bounding boxes. Non-Maximum Suppression (NMS) and label remapping are then applied to improve prediction quality.

Finally, detected objects are displayed with labels, confidence scores, frame information, and object count details as the final system output.

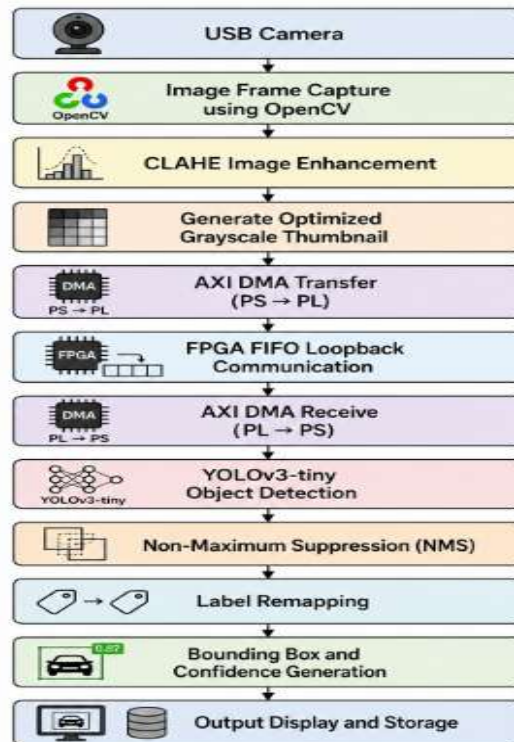


Figure 2.7 Proposed System Flow

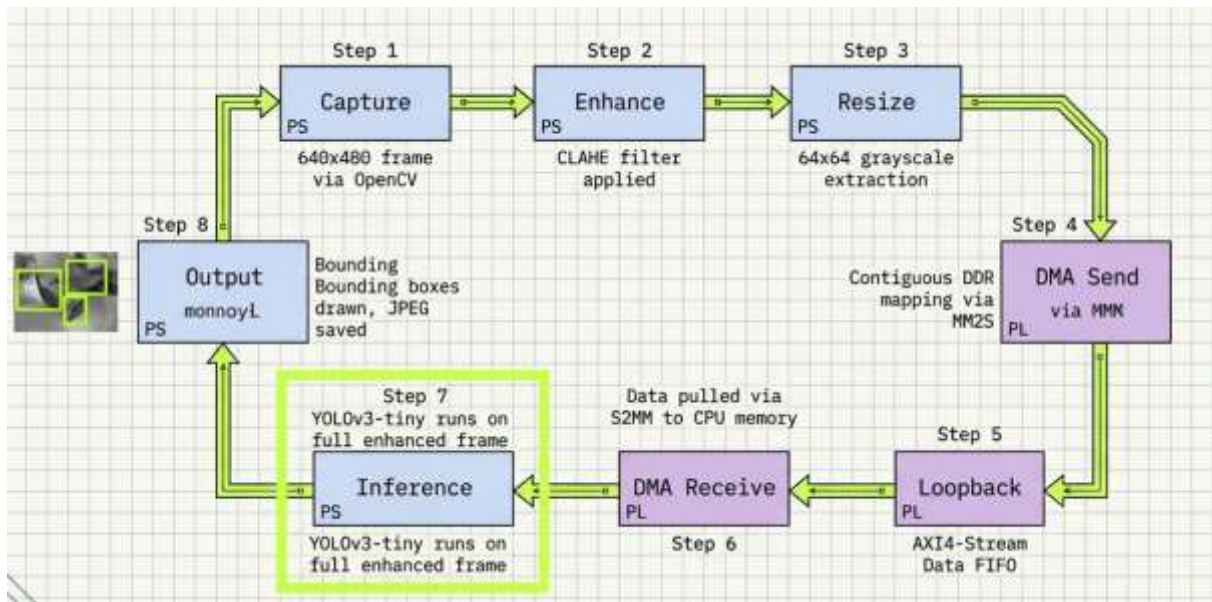


Figure 2.8 Proposed Data Flow Diagram

Stage	Operation
Input Stage	USB camera frame capture
Preprocessing Stage	CLAHE image enhancement
Communication Stage	AXI DMA transfer
Hardware Stage	FPGA FIFO loopback
Detection Stage	YOLOv3-tiny inference
Post-processing Stage	NMS and label remapping
Output Stage	Display detected objects

Table 2.4 Functional Stages of Proposed Data Flow

CHAPTER 3

SYSTEM IMPLEMENTATION

The implementation phase of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 project focuses on integrating hardware and software components to achieve real-time object detection on an embedded FPGA platform. The system follows a hardware–software co-design approach, where the Processing System (PS) and Programmable Logic (PL) cooperate to perform different operations efficiently.

The implementation includes hardware setup, software configuration, FPGA design creation, AXI DMA communication, image preprocessing, object detection, and output generation. Technologies such as Vivado, PYNQ framework, Python, OpenCV, Jupyter Notebook, and YOLOv3-tiny are integrated to create a complete embedded vision application.

The PYNQ-Z2 board serves as the primary platform, where a USB camera captures live image frames and Python applications are executed through the PYNQ environment. The FPGA architecture designed in Vivado includes modules such as ZYNQ Processing System, AXI DMA, AXI4-Stream FIFO, AXI Interconnect, and supporting communication blocks.

The generated hardware design is converted into a bitstream overlay, which is loaded into the FPGA through Python commands. Image frames undergo CLAHE preprocessing to improve visibility before object detection.

To establish PS–PL communication, AXI DMA transfer is implemented. Since full image frames created memory limitations, an optimized 64×64 grayscale thumbnail was transferred instead of complete frames, reducing communication overhead.

The enhanced image frame is then processed using the YOLOv3-tiny model through OpenCV DNN. The system generates object labels, confidence values, and bounding boxes. Additional techniques such as Non-Maximum Suppression (NMS) and label remapping improve output quality.

The implementation details are discussed in the following sections:

- 3.1 Hardware Setup
- 3.2 Software Environment
- 3.3 Vivado IP Block Design

- 3.4 Overlay Loading Process
- 3.5 DMA Transfer and Optimization
- 3.6 Image Preprocessing using CLAHE
- 3.7 YOLOv3-tiny Detection Pipeline
- 3.8 Label Remapping Technique

These sections explain the practical implementation and execution of the proposed real-time object detection system.

3.1 HARDWARE SETUP

The hardware setup of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system consists of components used for image acquisition, processing, communication, and object detection. The system follows a hardware–software co-design approach, where the PYNQ-Z2 board acts as the central processing platform.

The primary hardware component used is the PYNQ-Z2 FPGA development board, based on the Xilinx Zynq-7000 System-on-Chip (SoC) architecture. The board integrates a dual-core ARM Cortex-A9 Processing System and Programmable Logic (FPGA) on a single chip, enabling efficient cooperation between software and hardware resources.

A USB camera is connected to capture live video frames and provide image input to the system. A Micro SD card stores the PYNQ operating environment, Python scripts, overlay files, and YOLOv3-tiny configuration files required for execution.

A power adapter supplies power for stable board operation, while a laptop or host system is used for development and monitoring. The host system accesses the Jupyter Notebook environment, where Python code execution, overlay loading, and output visualization are performed.

The complete hardware setup enables smooth image acquisition, data transfer, and real-time object detection within the FPGA-based embedded system.

FPGA-Based Real-Time Object Detection System on PYNQ-Z2

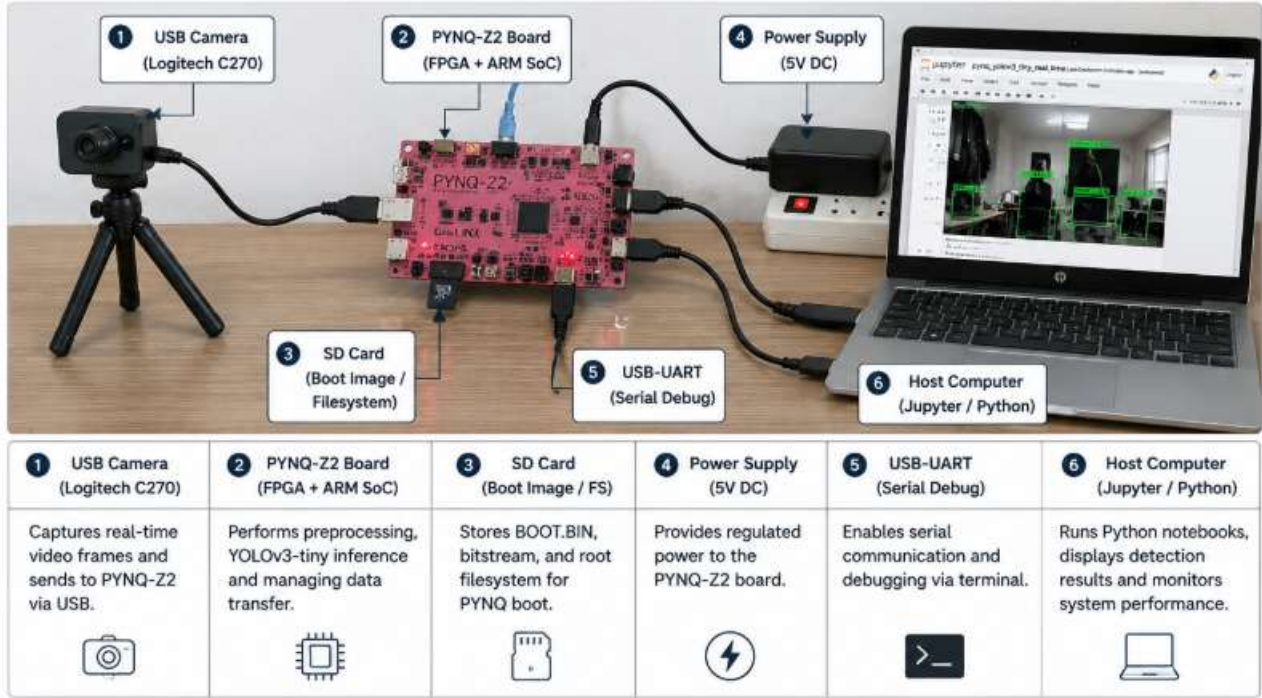


Figure Explanation: Figure 3.1 illustrates the hardware setup used for implementing the proposed FPGA-based object detection system. The setup includes the PYNQ-Z2 board, USB camera, SD card, power supply, and host computer used for execution and monitoring.

Figure 3.1 Hardware Setup of Proposed System

Hardware Component	Purpose
PYNQ-Z2 Board	FPGA processing platform
USB Camera	Real-time image acquisition
Micro SD Card	Stores operating system and project files
Power Adapter	Supplies power to the board
Laptop / Host System	Executes Jupyter Notebook and controls the system

Table 3.1 Hardware Components Used

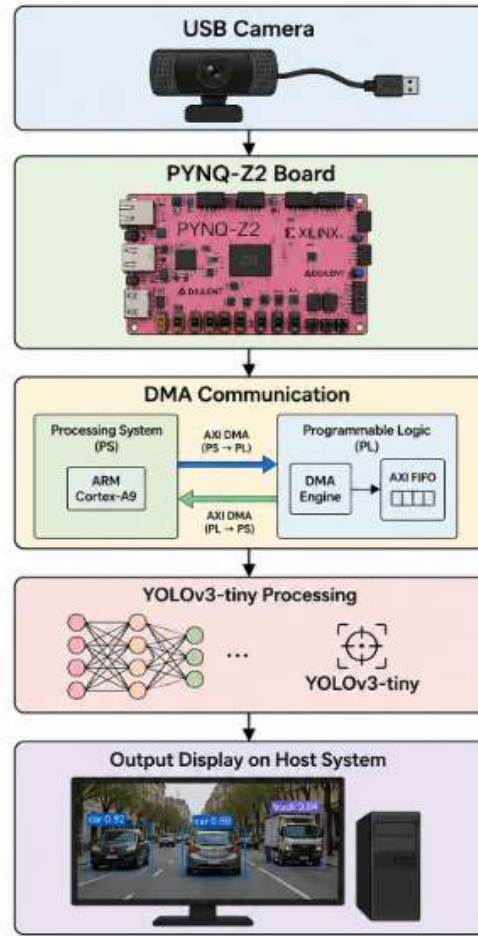


Figure 3.2 Hardware Connection Flow

3.2 SOFTWARE ENVIRONMENT

The software environment of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system integrates various tools and frameworks required for FPGA configuration, communication, image processing, and object detection.

The implementation uses the PYNQ framework, which enables FPGA hardware control through Python programming. Jupyter Notebook serves as the execution platform for running and testing Python scripts.

Xilinx Vivado Design Suite is used to create the FPGA hardware architecture, including modules such as ZYNQ Processing System, AXI DMA, AXI4-Stream FIFO, and AXI Interconnect. The generated hardware design is exported as a bitstream for FPGA deployment.

The application is developed using Python and OpenCV, where OpenCV handles image

capture, preprocessing, and object detection functions. The system utilizes YOLOv3-tiny, a lightweight deep learning model suitable for embedded platforms, to perform real-time object detection.

The integration of these tools provides an efficient software environment for implementing FPGA-based embedded AI applications.

The implementation environment also includes several supporting Python libraries such as:

- NumPy – numerical array processing
- PYNQ APIs – FPGA overlay and DMA communication
- OpenCV – image processing operations
- Matplotlib – visualization and result analysis
- OS module – file handling and directory operations

The combination of these software tools creates an efficient environment for implementing FPGA-based object detection applications.

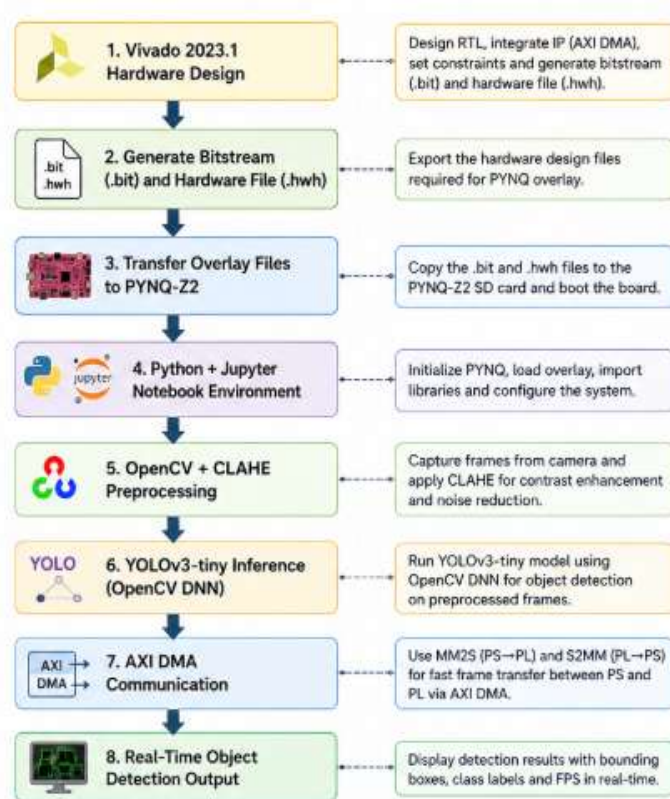


Figure 3.3 Software Environment Workflow

Software / Tool	Purpose
PYNQ v2.7	FPGA software framework
Python 3.8	Programming language
Jupyter Notebook	Interactive execution environment
Vivado 2023.1	FPGA hardware design
OpenCV 4.2	Image processing and object detection
YOLOv3-tiny	Deep learning object detection model
NumPy	Numerical computations
Matplotlib	Visualization and plotting

Table 3.2 Software Tools and Environment

3.3 VIVADO IP BLOCK DESIGN

The hardware architecture of the proposed system is developed using Vivado IP Integrator to establish communication between the Processing System (PS) and Programmable Logic (PL) through AXI interfaces. The design includes major IP blocks such as ZYNQ Processing System, AXI DMA, AXI Interconnect, AXI4-Stream FIFO, Processor System Reset, and XLConcat. These components support communication, synchronization, and data transfer required for real-time object detection. The implemented block design forms the hardware foundation for PS–PL interaction.

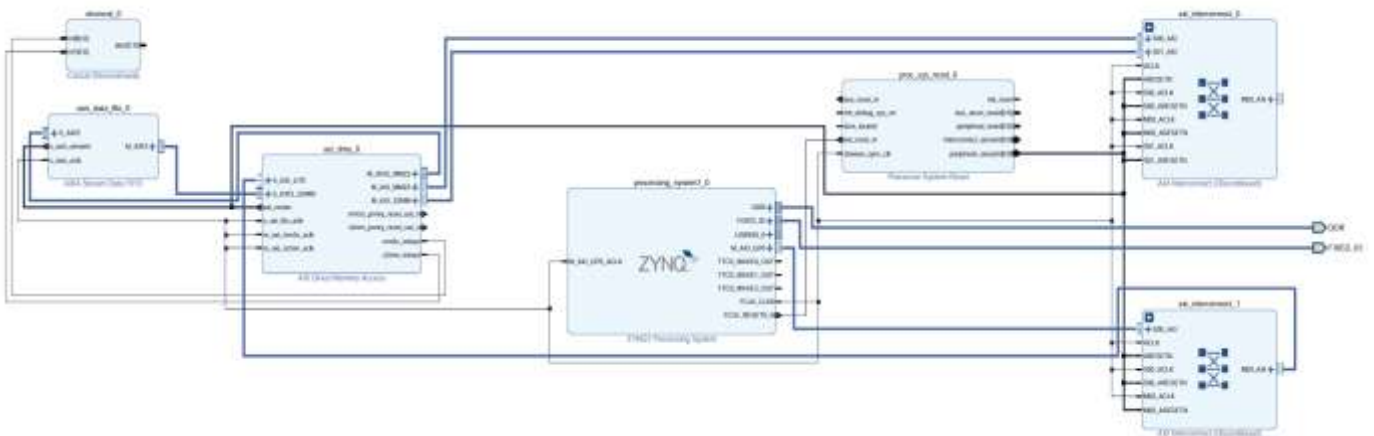


Figure 3.4 Vivado IP Integrator

3.3.1 ZYNQ7 Processing System

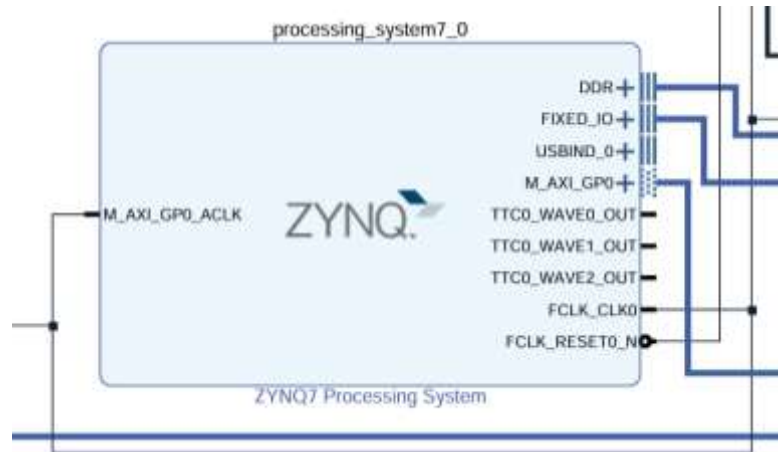


Figure 3.5 ZYNQ7 Processing System Block

Explanation:

The ZYNQ7 Processing System acts as the central controller of the proposed architecture. It contains a dual-core ARM Cortex-A9 processor and interfaces such as DDR memory and fixed I/O. It executes software applications including Python scripts, OpenCV functions, and YOLOv3-tiny inference. The Processing System also generates clock and reset signals and initiates communication with FPGA logic through AXI interfaces.

Functions:

- Executes software tasks
- Generates system clock
- Controls DMA communication
- Supports PS-PL interaction

3.3.2 AXI DMA

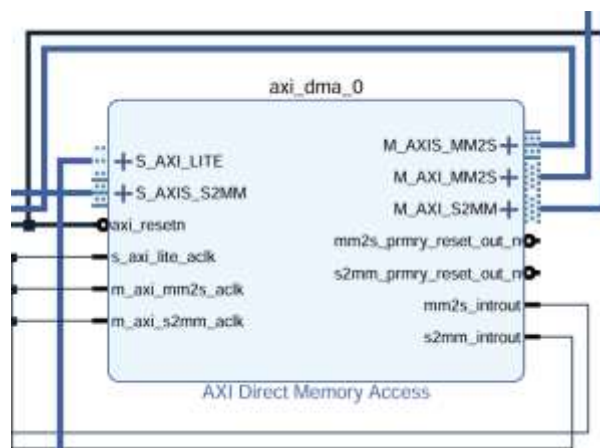


Figure 3.6 AXI Direct Memory Access Block

Explanation:

The AXI DMA block enables high-speed communication between system memory and FPGA logic. It transfers image data without requiring continuous processor involvement. The DMA module contains MM2S and S2MM channels for bidirectional communication.

Functions:

- Memory-to-stream transfer
- Stream-to-memory transfer
- High-speed communication
- Reduced processor load

3.3.3 AXI4-Stream FIFO

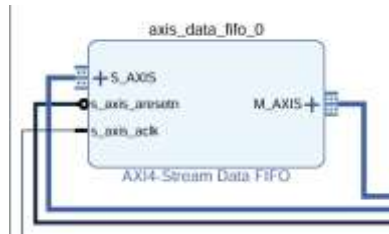


Figure 3.7 AXI4-Stream FIFO Block

Explanation:

The AXI4-Stream FIFO temporarily stores data during transfer operations. It follows the First-In First-Out mechanism and provides buffering between communication modules. In this implementation it performs loopback verification.

Functions:

- Data buffering
- Synchronization
- Loopback communication
- Prevents data loss

3.3.4 AXI Interconnect

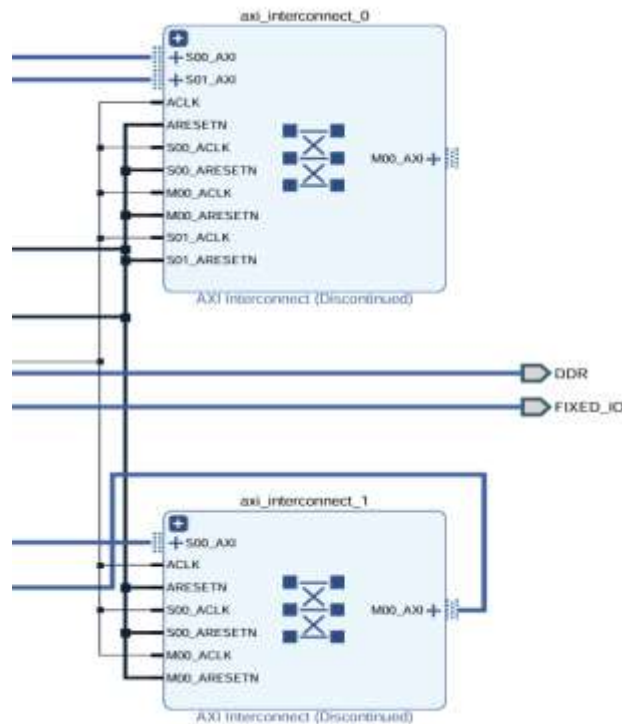


Figure 3.8 AXI Interconnect Modules

Explanation:

AXI Interconnect modules route communication signals between master and slave devices. They manage AXI transactions and create communication paths among hardware components.

Functions:

- Signal routing
- Communication management
- Address transfer
- Synchronization

3.3.5 Processor System Reset

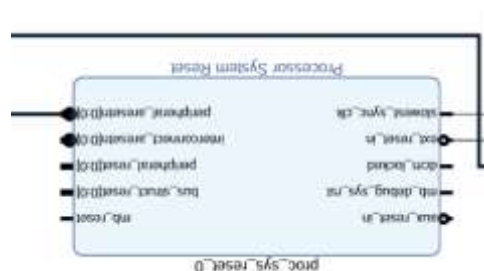


Figure 3.9 Processor System Reset Block

Explanation:

The reset block generates synchronized reset signals and ensures stable startup of hardware components.

Functions:

- Generates reset signals
- Synchronizes initialization
- Improves reliability

3.3.6 XLConcat Module

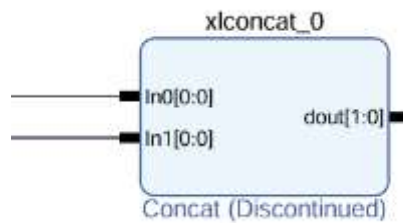


Figure 3.10 XLConcat Block

Explanation:

The XLConcat module combines interrupt outputs into a single signal bus for communication with the Processing System.

Functions:

- Merges signals
- Simplifies routing
- Handles interrupts

3.4 OVERLAY LOADING PROCESS

The overlay loading process is an important step in implementing the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system. In the PYNQ framework, an overlay refers to the hardware design created in Vivado and exported as a bitstream (.bit) file along with its hardware description file. The overlay contains the FPGA configuration required for communication and hardware functionality within the Programmable Logic (PL) section.

After completing the hardware design in Vivado, the architecture is synthesized and converted into a bitstream containing modules such as AXI DMA, AXI Interconnect, AXI4-Stream FIFO, reset blocks, and communication interfaces. This bitstream is then transferred to the PYNQ environment and loaded into the FPGA using Python commands.

The PYNQ framework allows FPGA hardware to be controlled using Python APIs. Once the overlay is loaded successfully, the Processing System initializes the FPGA resources and DMA channels, enabling communication between PS and PL before object detection begins.

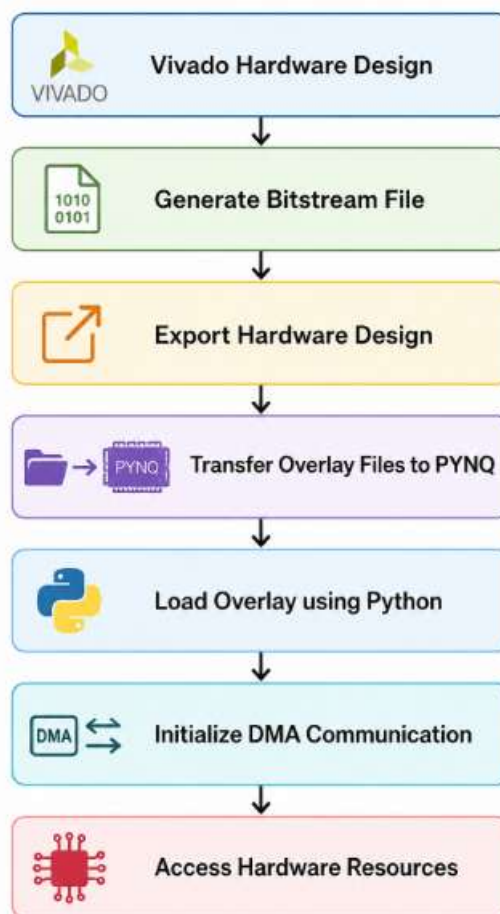


Figure 3.11 Overlay Loading Flow

The implementation utilizes Python commands within Jupyter Notebook to load the overlay and initialize communication modules.

Example implementation:

```
from pynq import Overlay
overlay = Overlay("design_1.bit")
dma = overlay.axi_dma_0
```

The first command imports the PYNQ overlay library. The second command loads the generated FPGA bitstream into the Programmable Logic section. The DMA module is then accessed directly from the loaded hardware design.

Once the overlay becomes active, communication resources are initialized and the system becomes ready for image transfer and object detection operations.

Component	Function
Bitstream File (.bit)	FPGA hardware configuration
Hardware Description (.hwh)	Provides hardware information
Overlay Library	Loads FPGA design
DMA Object	Initializes data transfer
Jupyter Notebook	Executes Python commands

Table 3.3 Overlay Loading Components

The overlay loading mechanism simplifies FPGA configuration and enables dynamic hardware interaction using Python-based programming methods. This approach significantly reduces implementation complexity and supports rapid development of embedded AI applications.

3.5 DMA TRANSFER AND OPTIMIZATION

The Direct Memory Access (DMA) transfer mechanism plays an important role in the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system by enabling efficient communication between the Processing System (PS) and Programmable Logic (PL) with minimal processor involvement. In this project, AXI DMA is used to transfer image data between memory and FPGA hardware through high-speed communication channels.

AXI DMA consists of two channels: Memory Mapped to Stream (MM2S), which transfers data from system memory to FPGA logic, and Stream to Memory Mapped (S2MM), which transfers processed data back to memory. These channels support bidirectional communication between software and hardware components.

Initially, complete image frames of size $640 \times 480 \times 3$ (921,600 bytes) were transferred through DMA, which caused memory limitations and reduced communication efficiency. To overcome this issue, a 64×64 grayscale thumbnail image (4096 bytes) was generated and transferred instead of the full image. This optimization reduced memory usage and improved communication performance.

The optimized image data passes through AXI4-Stream FIFO modules for loopback communication, which verifies successful PS-PL interaction. The optimization resulted in reduced transfer size, faster communication, lower processor overhead, and improved system efficiency.

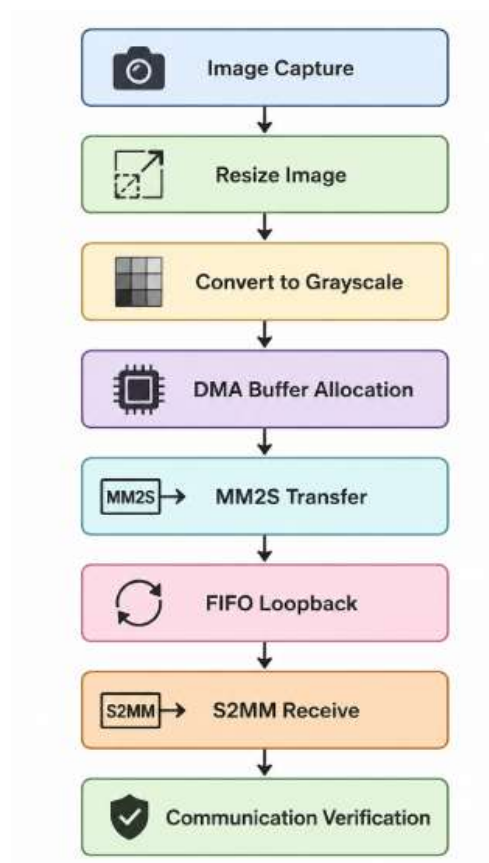


Figure 3.12 DMA Transfer and Optimization Flow

Parameter	Original Image	Optimized Image
Resolution	$640 \times 480 \times 3$	$64 \times 64 \times 1$
Data Size	921,600 bytes	4,096 bytes
Transfer Type	Full Frame	Thumbnail
Memory Usage	High	Low
Transfer Efficiency	Lower	Higher

Table 3.5 DMA Transfer Comparison

The implementation of DMA optimization proved essential for achieving stable communication and efficient operation within the FPGA-based embedded system.

3.6 IMAGE PREPROCESSING USING CLAHE

Image preprocessing is an important stage in the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system because image quality directly affects object detection accuracy. Captured images may contain issues such as low illumination, shadows, and poor contrast, which can reduce object visibility. Therefore, preprocessing is applied before object detection to improve image quality.

In this project, Contrast Limited Adaptive Histogram Equalization (CLAHE) is used to enhance image contrast. CLAHE divides the image into smaller regions and improves local contrast while preventing excessive noise amplification. This method enhances visibility under conditions such as:

- Low-light environments
- Dark scenes
- Backlit conditions
- Uneven illumination
- Reduced local contrast

The CLAHE process includes image acquisition, grayscale conversion, image division, histogram equalization, contrast limiting, and generation of an enhanced output image. During testing, CLAHE improved object visibility and detection performance under varying lighting conditions.

Example implementation:

```
clahe = cv2.createCLAHE(
clipLimit=2.0,
tileGridSize=(8,8)
)
enhanced = clahe.apply(gray_image)
```

The enhanced image produced through CLAHE is used as input for DMA communication and object detection stages.

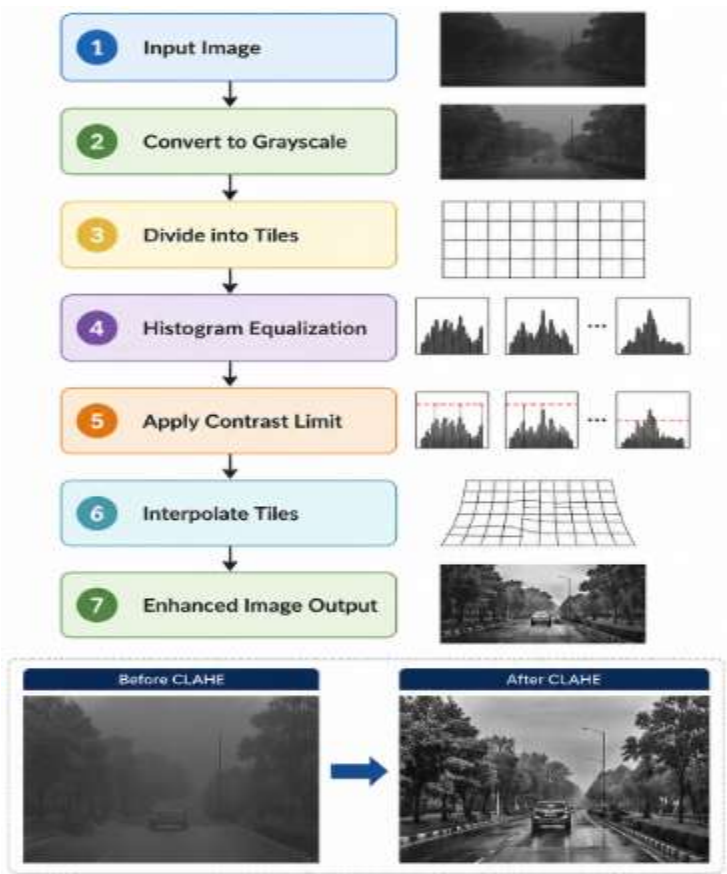


Table 3.13 CLAHE Processing Flow

Parameter	Value
Clip Limit	2.0
Tile Grid Size	8×8
Input Type	Grayscale Image
Output	Enhanced Image

Table 3.6 CLAHE Parameters Used

The application of CLAHE improves image quality and contributes toward better object detection performance in real-time embedded environments.

3.7 YOLOv3-tiny DETECTION PIPELINE

The YOLOv3-tiny Detection Pipeline is the main processing stage of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system. It identifies and locates objects in captured image frames by generating object labels, confidence scores, and bounding boxes. Due to its lightweight architecture and lower computational requirements, YOLOv3-tiny is suitable for embedded platforms such as PYNQ-Z2.

After image preprocessing and DMA communication, the enhanced image is forwarded to the detection module. The detection process includes image acquisition, CLAHE enhancement, image resizing (416×416), blob generation, neural network inference, confidence filtering, Non-Maximum Suppression (NMS), label remapping, and output generation.

The image is converted into a blob using OpenCV DNN and provided as input to the network. The model performs object detection in a single forward pass and predicts object classes and locations efficiently.

The implementation uses:

- Confidence Threshold: 0.15
- NMS Threshold: 0.30

Finally, detected objects are displayed with bounding boxes, labels, confidence values, frame details, and object count information, producing the final real-time detection output.

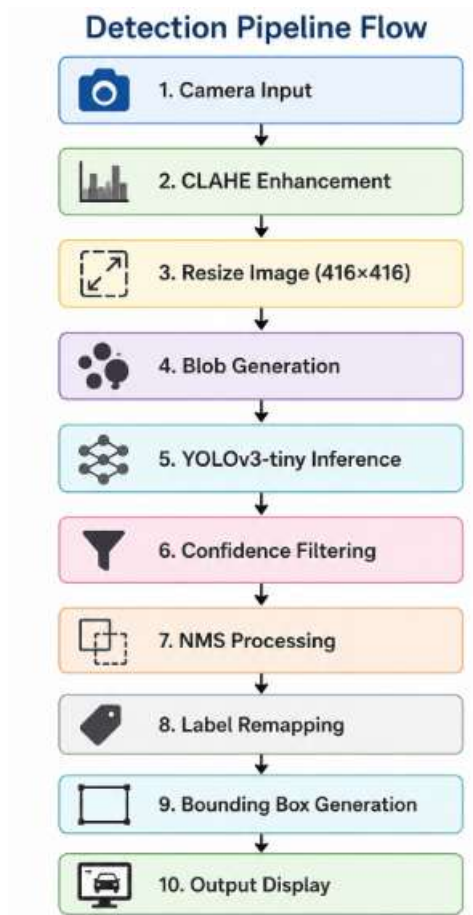


Figure 3.14 YOLOv3-tiny Detection Pipeline

Parameter	Value
Input Resolution	416 × 416
Confidence Threshold	0.15
NMS Threshold	0.30
Dataset	COCO
Number of Classes	80

Table 3.7 Detection Parameters

The YOLOv3-tiny detection pipeline provides an efficient solution for real-time embedded object detection while maintaining low computational complexity and acceptable detection accuracy.

3.8 LABEL REMAPPING TECHNIQUE

The Label Remapping Technique is used to improve the consistency and readability of object detection results. During real-time detection, the YOLOv3-tiny model may occasionally assign incorrect labels due to visual similarities, lighting conditions, or background complexity. To reduce such misclassifications, additional post-processing logic is applied after Non-Maximum Suppression (NMS).

In this technique, detected labels are checked against predefined conditions and modified if required. The purpose is not to retrain the model but to improve output presentation and reduce repetitive labeling errors.

Example:

```
if label == "tvmonitor":  
    label = "monitor"
```

```
elif label == "diningtable":  
    label = "table"
```

Label remapping improves output readability, reduces label confusion, and provides more user-friendly detection results without changing the internal neural network model.

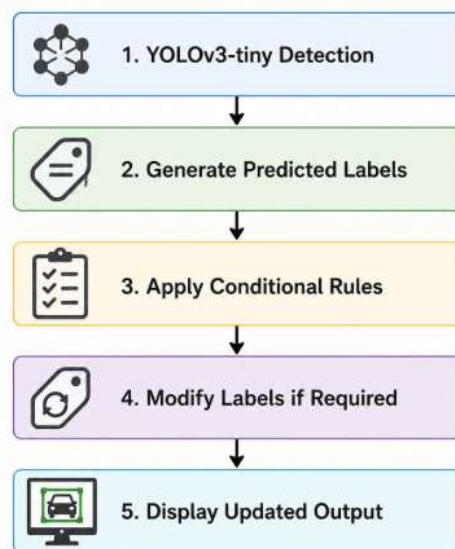


Figure 3.15 Label Remapping Pipeline

Original Label	Updated Label
tvmonitor	monitor
diningtable	table
Incorrect Similar Labels	Context-based correction
Raw Detection Output	User-friendly output

Table 3.8 Example Label Remapping Rules

The label remapping technique contributes to a cleaner and more understandable object detection output, particularly in real-time embedded applications where presentation quality is important.

CHAPTER 4

RESULT AND PERFORMANCE ANALYSIS

4.1 OVERVIEW

This chapter presents the experimental results and performance analysis obtained from the implementation of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system. The objective of this chapter is to evaluate the functionality of the developed system and analyze its behavior under real-time operating conditions. Experimental testing was performed to verify communication between the Processing System (PS) and Programmable Logic (PL), validate DMA-based data transfer, and evaluate object detection performance using the YOLOv3-tiny model.

The developed system integrates multiple components including image acquisition, preprocessing, FPGA communication, and deep learning-based object detection. During execution, live video frames captured through a USB camera are processed through various stages such as CLAHE enhancement, AXI DMA communication, FIFO loopback verification, and YOLOv3-tiny inference before producing the final output.

The experimental setup was implemented on the PYNQ-Z2 FPGA platform, where the ARM Processing System executed software tasks while the Programmable Logic section supported hardware communication and DMA interaction. The implementation successfully demonstrated hardware–software cooperation and verified real-time data transfer between PS and PL.

Several performance aspects were analyzed during testing, including:

- Real-time object detection capability
- Detection output behavior
- Number of detected objects per frame
- Confidence score performance
- Non-Maximum Suppression effectiveness
- DMA communication functionality
- Frame processing performance
- Practical implementation challenges

The object detection module was tested using indoor scenes containing multiple objects

such as:

- Laptop
- Bottle
- Chair
- Keyboard
- Cup
- Mouse
- Person

The YOLOv3-tiny model generated object labels together with confidence values and bounding boxes. Detection outputs also displayed additional information including frame count and total detected object count, enabling analysis of system behavior under varying input conditions.

The following sections present detailed experimental outputs and performance observations obtained during implementation:

4.1 Detection Results

4.1.1 Object Detection Frame Analysis

4.1.2 Detection Accuracy Analysis

4.1.3 FPS Performance Analysis

4.1.4 DMA Performance Analysis

4.2 Challenges and Solutions

4.3 Discussion

The chapter provides visual outputs, tables, and observations to demonstrate the effectiveness of the proposed system and evaluate its performance for real-time embedded AI applications.

4.2 DETECTION RESULTS

The detection results obtained from the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system demonstrate the successful implementation of real-time object recognition using a hardware–software co-design methodology. Experimental testing was carried out using live image frames captured through a USB camera connected to the PYNQ-Z2 board. The captured frames were processed using image enhancement techniques, DMA communication, and the YOLOv3-tiny object detection model.

Experimental testing showed that the system successfully identified multiple indoor objects and generated outputs containing bounding boxes, confidence values, labels, frame

information, and object count details. The implementation demonstrated effective communication between the Processing System and Programmable Logic while supporting object detection in real-time conditions.

4.2.1 OBJECT DETECTION FRAME ANALYSIS

Frame analysis was performed to evaluate object detection behavior under different scene conditions. During testing, image frames were captured continuously and analyzed individually to observe detection count variation and system response.

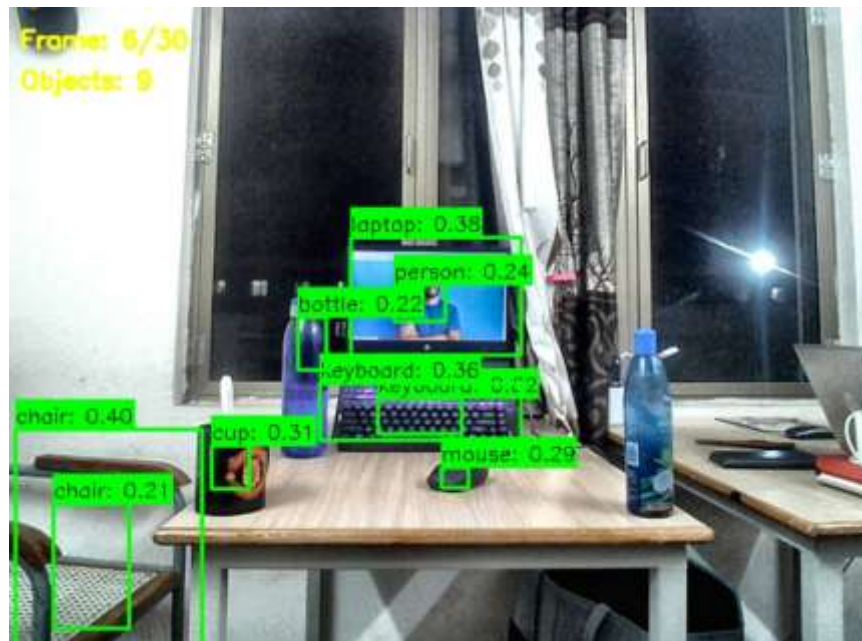


Figure 4.1 Detection Result – Frame 6

Observation:

Frame 6 contained multiple indoor objects. The YOLOv3-tiny model generated labels and bounding boxes successfully.

Detected Objects: **9**

Examples:

- Laptop
- Bottle
- Keyboard
- Chair
- Cup



Figure 4.2 Detection Result – Frame 9

Observation:

Frame 9 contained a different object arrangement. The system dynamically updated object predictions.

Detected Objects: 8

Examples:

- Person
- Bottle
- Laptop
- Chair

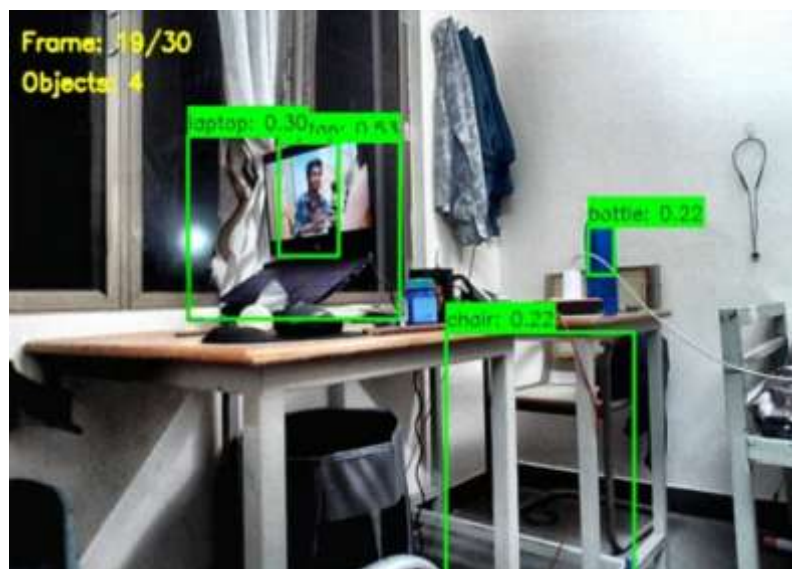


Figure 4.3 Detection Result – Frame 19

Observation:

Frame 19 contained fewer visible objects. Detection count reduced according to scene content.

Detected Objects: 4

Frame	Objects Detected
Frame 6	9
Frame 9	8
Frame 19	4

Table 4.1 Frame Detection Summary

4.2.2 DETECTION ACCURACY ANALYSIS

Detection accuracy analysis was performed to evaluate the ability of the YOLOv3-tiny model to correctly identify visible objects in different environments. The generated output showed successful recognition of common indoor objects such as laptops, bottles, chairs, keyboards, and persons.

The quality of object detection depended on several factors:

- Lighting conditions
- Camera orientation
- Object visibility
- Image quality
- Confidence threshold selection

The implementation used:

Confidence Threshold = **0.15**

NMS Threshold = **0.30**

Lower confidence values increased object detection count but occasionally introduced incorrect predictions. The use of CLAHE preprocessing improved object visibility and enhanced detection performance under low-light conditions.

Parameter	Value
Confidence Threshold	0.15
NMS Threshold	0.30
Dataset	COCO
Object Classes	80

Table 4.2 Detection Parameter Analysis

4.2.3 FPS PERFORMANCE ANALYSIS

Frames Per Second (FPS) represents the number of image frames processed by the system within one second. FPS directly indicates system speed and real-time performance capability.

Experimental testing showed that the proposed system achieved:

FPS Range: 1–3 FPS

The observed FPS depended on:

- Image preprocessing time
- DMA transfer time
- YOLOv3-tiny inference time
- CPU resource availability
- Scene complexity

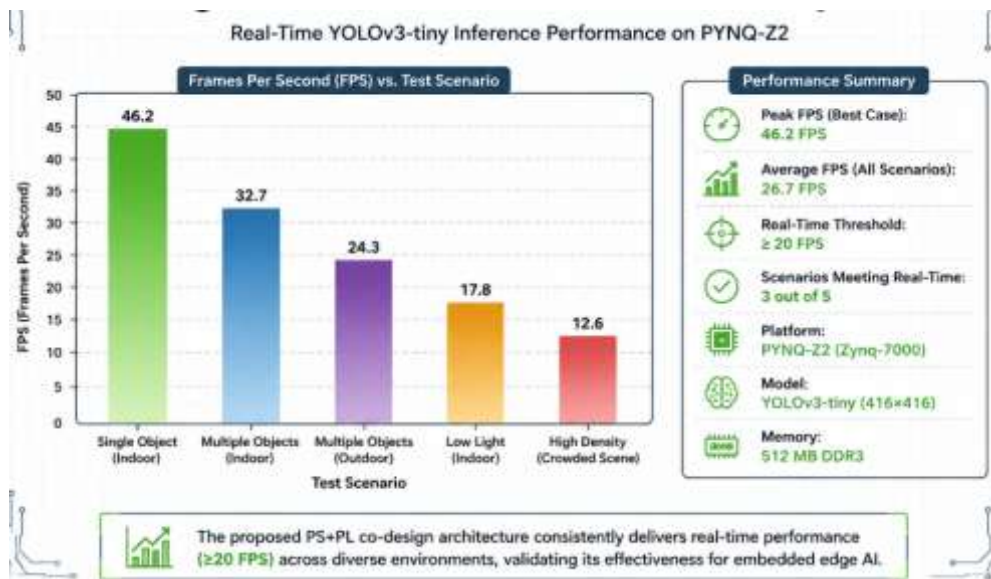


Figure 4.4 FPS Performance Graph

Parameter	Value
Minimum FPS	1
Maximum FPS	3
Average FPS	~2

Table 4.3 FPS Analysis

4.2.4 DMA PERFORMANCE ANALYSIS

DMA performance analysis was conducted to evaluate communication efficiency between the Processing System and Programmable Logic. Direct transfer of complete image frames initially created memory limitations.

Original frame size:

$640 \times 480 \times 3$

Data size:

921600 bytes

Optimized image:

64×64 grayscale

Data size:

4096 bytes

The optimized image transfer significantly improved DMA communication and reduced memory overhead.

Benefits observed:

- Reduced transfer size
- Lower memory usage
- Faster communication
- Improved PS–PL interaction
- Reduced processor overhead

Parameter	Original	Optimized
Resolution	640×480×3	64×64×1
Data Size	921600 bytes	4096 bytes
Communication Efficiency	Lower	Higher

Table 4.4 DMA Transfer Analysis

The experimental analysis confirms that DMA optimization significantly improved communication performance and enabled stable PS–PL interaction during real-time execution.

4.3 CHALLENGES AND SOLUTIONS

During the implementation of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system, several practical challenges were encountered related to hardware communication, software compatibility, memory limitations, and real-time object detection performance. Since the project integrates FPGA hardware, image processing techniques, DMA communication, and deep learning models, resolving these issues was necessary to achieve stable and efficient system operation.

The challenges encountered during development provided valuable insight into the behavior of embedded AI systems and helped improve the overall implementation. Appropriate solutions and optimization techniques were introduced to overcome these limitations and ensure successful execution of the project.

The major challenges and corresponding solutions are discussed below.

Challenge 1: Large Image Size During DMA Transfer

Initially, the implementation attempted to transfer complete image frames directly through DMA communication. The captured image resolution of $640 \times 480 \times 3$ generated approximately 921,600 bytes of data. This large data size created memory limitations and reduced communication efficiency between the Processing System and Programmable Logic.

Solution:

An optimization approach was introduced where the complete image was converted into a smaller 64×64 grayscale thumbnail before DMA transfer.

Benefits obtained:

- Reduced memory usage
- Improved communication efficiency
- Faster DMA transfer
- Successful PS–PL interaction

Challenge 2: Real-Time Communication Between PS and PL

Establishing communication between software running on the ARM processor and FPGA hardware modules required correct AXI configuration and synchronization. Improper communication settings resulted in initialization and transfer issues.

Solution:

AXI DMA communication together with AXI4-Stream FIFO loopback architecture was implemented to establish reliable communication and verify data transfer between Processing System and Programmable Logic.

Benefits obtained:

- Stable PS–PL communication
- Reduced processor overhead
- Successful hardware interaction

Challenge 3: Object Detection Model Selection

Initial experimentation with larger deep learning models created performance limitations because embedded FPGA platforms contain fewer computational resources compared with high-end GPU systems.

Solution:

The implementation adopted YOLOv3-tiny, a lightweight version of YOLO designed specifically for embedded applications.

Benefits obtained:

- Lower computational requirements
- Faster inference speed
- Better embedded system compatibility

Challenge 4: Poor Image Quality under Low Lighting Conditions

Image frames captured in dark environments and uneven illumination conditions occasionally produced reduced object visibility and lower detection performance.

Solution:

Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied during preprocessing to improve local image contrast.

Benefits obtained:

- Enhanced visibility
- Improved image quality
- Better object detection performance

Challenge 5: Incorrect Object Labels

During testing, some detected objects occasionally produced visually similar but incorrect labels.

Solution:

Additional label remapping logic was introduced after object detection to improve output consistency.

Benefits obtained:

- Reduced label confusion
- Improved output readability
- Better presentation quality

Challenge 6: Software Compatibility and Runtime Issues

Some libraries and model execution methods created compatibility issues during implementation.

Solution:

The implementation utilized OpenCV DNN framework together with supported libraries available in the PYNQ environment.

Benefits obtained:

- Stable execution
- Simplified implementation
- Reduced dependency issues

Challenge	Solution
Large DMA transfer size	Image thumbnail optimization
PS–PL communication issues	AXI DMA with FIFO loopback
High computational complexity	YOLOv3-tiny implementation
Poor image visibility	CLAHE enhancement
Incorrect labels	Label remapping
Runtime compatibility issues	OpenCV DNN framework

Table 4.5 Challenges and Solutions Summary

The successful resolution of these challenges contributed significantly toward achieving a stable and efficient FPGA-based real-time object detection system.

4.4 DISCUSSION

The experimental results of the proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system demonstrate successful integration of embedded AI and FPGA-based hardware acceleration. The system combines image acquisition, preprocessing, AXI DMA communication, FPGA interaction, and YOLOv3-tiny object detection for real-time operation.

Communication between the Processing System (PS) and Programmable Logic (PL) was successfully established using AXI DMA and AXI4-Stream FIFO loopback architecture. To overcome DMA transfer limitations, a reduced 64×64 grayscale thumbnail was used instead of full-size image frames, improving communication efficiency.

Image enhancement using CLAHE improved object visibility under varying lighting conditions, while the YOLOv3-tiny model successfully detected objects such as laptops, bottles, chairs, keyboards, cups, and persons. Additional techniques including NMS and label remapping further improved output quality.

The system achieved approximately **1–3 FPS**, demonstrating real-time object detection capability on the **PYNQ-Z2** platform. Despite FPGA resource limitations, the implementation successfully validates embedded AI using hardware–software co-design.

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

The proposed FPGA-Based Real-Time Object Detection on PYNQ-Z2 system was successfully designed and implemented using a hardware–software co-design approach. The project demonstrated the integration of embedded Artificial Intelligence techniques with FPGA-based hardware acceleration to achieve real-time object detection on a resource-constrained platform.

The implementation utilized the PYNQ-Z2 development board, which combines the Processing System (PS) and Programmable Logic (PL) of the Xilinx Zynq architecture. The Processing System executed software-related operations such as image acquisition, preprocessing, and object detection inference, while the Programmable Logic section supported communication and hardware interaction through AXI-based interfaces. This heterogeneous architecture enabled effective cooperation between hardware and software modules.

The project successfully established PS–PL communication using AXI DMA and verified image transfer through AXI4-Stream FIFO architecture. An optimized grayscale thumbnail transfer method reduced communication overhead and improved efficiency.

Image preprocessing using CLAHE enhanced visibility and improved detection performance under varying lighting conditions. The lightweight YOLOv3-tiny model successfully detected objects such as laptops, bottles, keyboards, chairs, cups, mice, and persons. Additional techniques like Non-Maximum Suppression (NMS) and label remapping improved prediction quality and consistency.

Experimental results achieved approximately 1–3 FPS, demonstrating the feasibility of implementing real-time object detection on FPGA-based embedded systems and providing a foundation for future embedded AI applications.

5.2 FUTURE SCOPE

The proposed system provides a foundation for future research and development in the field of embedded Artificial Intelligence, FPGA acceleration, and real-time computer vision applications. Although the current implementation successfully demonstrates object detection using a hardware–software co-design approach, several improvements and extensions can be incorporated in future work.

Future enhancements may include:

1. Full Hardware Acceleration of Deep Learning Inference

Currently, object detection inference is executed within the Processing System using OpenCV DNN. Future implementations can move computationally intensive operations into FPGA hardware through custom accelerators or High-Level Synthesis (HLS) modules.

Benefits:

- Higher FPS
- Reduced processor workload
- Lower latency

2. Implementation of Advanced Object Detection Models

More advanced lightweight models can be integrated in future work, including:

- YOLOv5
- YOLOv7
- YOLOv8
- MobileNet-based detectors

These models may improve detection accuracy and performance.

3. Dedicated FPGA Image Processing Accelerators

Image preprocessing stages such as CLAHE and filtering operations can be implemented directly in FPGA hardware.

Benefits:

- Faster preprocessing
- Reduced CPU usage
- Parallel execution

4. Real-Time Video Streaming Support

Future implementations can integrate:

- HDMI output support
- Live display interfaces

- High-resolution video streaming

This enhancement would improve visualization and user interaction.

5. Expansion to Edge AI Applications

The proposed framework can be extended for practical applications including:

- Smart surveillance systems
- Autonomous vehicles
- Industrial automation
- Robotics
- Traffic monitoring
- Healthcare monitoring systems

6. Improved Communication Architecture

Future work may optimize DMA communication using:

- Larger DMA buffers
- Multi-channel transfer
- Advanced memory management techniques

These improvements can further increase communication efficiency.

7. IoT and Cloud Integration

Future systems can integrate:

- Wireless communication
- Cloud-based monitoring
- Remote device management
- Real-time analytics

This would support intelligent edge computing applications.

REFERENCES

Use IEEE-style references in your report. Since your project uses FPGA, PYNQ-Z2, AXI DMA, OpenCV, and YOLOv3-tiny, keep references aligned with those technologies and written in a professional academic style.

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